Steatocystoma Multiplex (MONDO:0008485): A Comprehensive Disease Characteristics Report

Causal gene: *KRT17* (keratin 17; HGNC:6427; NCBI Gene 3872; OMIM gene 148069). Disease OMIM 184500. Synonym: sebocystomatosis.

Summary

Steatocystoma multiplex (SM) is an autosomal dominant benign genodermatosis of the pilosebaceous unit caused by heterozygous missense mutations in the **1A helix-initiation rod domain of *KRT17* (keratin 17)**. The disorder manifests as multiple firm, non-tender, sebum-filled dermal cysts of the sebaceous follicle, distributed preferentially on the trunk, axillae, groin, and limbs, typically emerging around puberty under androgenic drive. The mutations cluster at a conserved helix-initiation motif, with **Arg94 (R94) as the predominant hotspot** (R94C, R94H) and **Asn92 (N92) as a secondary hotspot** (N92H, N92S). These variants act through a **dominant-negative mechanism**: the mutant keratin protein is incorporated into the keratin intermediate-filament network, where it disrupts hydrophobic interactions and surface-charge contacts required for keratin dimer/tetramer assembly, producing defective filament formation and cytoplasmic keratin aggregation in the cells lining the sebaceous duct.

A central and well-documented feature of this gene is **allelism**: identical *KRT17* mutations (notably R94C) can produce either classical SM or pachyonychia congenita type 2 (PC-2, now PC-K17), establishing that SM and PC-2 lie on a single *KRT17* phenotypic spectrum and that the specific phenotype is "independent of the specific mutation involved" (Covello 1998). Cutaneous cysts and natal teeth are the KRT17-signature phenotypes among the pachyonychia-congenita keratin genes. A systematic review of 145 confirmed cases provides quantitative natural-history data: median onset age of **23 years**, median diagnosis age of **31 years** (≈8-year diagnostic delay), with familial cases (~26%) showing significantly earlier onset and greater severity.

There is **no cure**; management is symptomatic and includes surgical excision, drainage, cryotherapy, oral isotretinoin (especially for the suppurative variant), laser, and biologics for inflammatory/suppurative disease. Because SM belongs to the dominant-negative keratinopathy class, **allele-specific siRNA** — clinically validated in pachyonychia congenita (TD101 against KRT6A, first-in-human phase Ib trial) — represents a rational, mechanism-targeted future therapy. SM does not affect lifespan and has no malignant potential; its burden is cosmetic, psychosocial, and related to secondary infection/scarring.

Key Findings

Finding 1 — SM is caused by heterozygous missense mutations in the 1A rod domain of *KRT17*

Landmark sequencing studies have established that SM arises from co-segregating heterozygous missense mutations clustered in the conserved **helix-initiation (1A) domain** of *KRT17*, inherited in an autosomal dominant manner. Smith et al. (1997) first reported the N92H and R94H mutations in two families diagnosed as steatocystoma multiplex: *"We also show heterozygous missense mutations in K17 (N92H and R94H) in two families diagnosed as steatocystoma multiplex"* (PMID: 9008238). Covello et al. (1998) extended the mutational spectrum with R94C and N92S and underlined the appendageal expression of K17: *"Keratin 17 (K17) is a differentiation-specific keratin expressed in the nail bed, hair follicle, sebaceous gland and other epidermal appendages"* (PMID: 9767294). The recurrent R94C allele was subsequently confirmed in a four-generation Chinese pedigree in which direct sequencing *"revealed a heterozygous mutation, c.280C/T (R94C), which was located in exon 1 of the KRT17 gene in all 10 affected family members"* — absent in 15 unaffected relatives and 100 controls (PMID: 26165312). Together these studies demonstrate strong genotype co-segregation and establish the 1A domain hotspot as the molecular signature of SM.

Finding 2 — SM and pachyonychia congenita type 2 are allelic *KRT17* disorders; phenotype is mutation-independent

A defining insight into SM genetics is that it is **allelic** with pachyonychia congenita type 2. Covello et al. reported two unrelated families carrying the **identical R94C mutation**, one presenting as classical PC-2 and the other as steatocystoma multiplex, and concluded that *"the alternate phenotypes which arise from these genetic lesions in K17 are independent of the specific*

mutation involved" (PMID: 9767294). This implies that **modifier genes and/or environmental factors**, rather than the specific nucleotide change, determine whether a carrier develops predominantly cysts (SM) or predominantly nail/palmoplantar disease (PC-2). More recently, Koprulu et al. (2022) showed that KRT17 R94H in the **homozygous (recessive)** state causes an atypical pachyonychia congenita, while *"Heterozygous variants in KRT17 cause PC2 (PC-K17) with main characteristics of pachyonychia, subungual keratosis, palmoplantar keratoderma, hyperhidrosis, oral leukokeratosis and epidermal cysts, or steatocystoma multiplex, both with dominant inheritance"* (PMID: 35676340). SM should therefore be understood as one pole of a continuous KRT17 phenotypic spectrum.

Finding 3 — Keratin 17 biology (hair-follicle cycling, stress response, TNFα/NF-κB) informs SM pathophysiology

K17 is not merely a structural scaffold; it is a stress-responsive, signaling-active intermediate filament. Tong & Coulombe (2006) showed that **Krt17-null mice develop alopecia** in the first postnatal week: *"Keratin 17 (K17)-null mice develop alopecia in the first week post-birth, correlating with hair shaft fragility and untimely apoptosis in the hair bulb"* (PMID: 16702408). They further identified *"K17 and TNFalpha as two novel and interdependent regulators of hair cycling,"* linking K17 to TNFα/NF-κB signaling through its interaction with TRADD; TNFα ablation partially rescues the hair-cycle defect. Additional work shows K17 regulates nuclear morphology and chromatin organization (PMID: 33008845) and participates in a Rac1-mediated DNA-damage response in keratinocytes (PMID: 41880216). These functions explain why K17 dysfunction preferentially perturbs appendageal (sebaceous gland, hair follicle, nail bed) homeostasis — the tissues in which cysts and dystrophy arise.

Finding 4 — SM clinical phenotype: benign sebum-filled sebaceous-follicle cysts with onset in adolescence/early adulthood

Clinically, steatocystomas are *"benign, sebum-filled cysts of the sebaceous follicle that typically manifest as firm nontender lesions of variable size and location"* (PMID: 41363957). They occur as isolated (simplex), grouped multiple (multiplex), or suppurative (steatocystoma multiplex suppurativum, SMS) forms. The characteristic distribution is described by Marasca et al. (2020): *"The occurrence of multiple asymptomatic cutaneous cysts on the axillae, groin, trunk, and limbs characterizes the disease"* (PMID: 33088817). The **histopathology is diagnostic**: a dermal cyst lined by stratified squamous epithelium with a characteristic eosinophilic corrugated ("shark-tooth") cuticle and **absent granular layer**, with flattened sebaceous gland lobules in or adjacent to the cyst wall. Sharma et al. describe *"a cyst… in the mid-dermis, lined*

by stratified squamous, agranular epithelium, which contained degenerated keratin" (PMID: 29998099). Lesions are usually asymptomatic but can be disfiguring and may become secondarily infected or scarred.

Finding 5 — Quantitative epidemiology: median onset age 23, familial cases ~26% with earlier onset, Arg94 predominant mutation site

A systematic review of 145 confirmed SM cases from 82 publications (Sun 2026) provides the most robust quantitative natural history to date. There is no clear geographical or ethnic predilection. *"The median age at onset was 23 years and the median age at diagnosis was 31 years, reflecting a diagnostic delay of approximately 8 years"* (PMID: 42518883). Familial disease is a meaningful minority with a more severe course: *"Familial cases (26.2%) had a significantly earlier age at onset (13.5 vs. 30.0 years, P < 0.001) and diagnosis (22.0 vs. 35.0 years, P < 0.001), and higher rates of infected lesions and nail dystrophy."* Hidradenitis suppurativa co-occurrence was more prevalent in females (12.5% vs 1.5%, P = 0.019). At the molecular level, *"KRT17 was the most frequently mutated gene, with Arg94 as the predominant mutation site."* These data confirm young-adult onset, a substantial diagnostic delay, and the Arg94 hotspot across aggregated cases.

Finding 6 — SM belongs to the dominant-negative keratinopathy class; allele-specific siRNA is a validated therapeutic paradigm

SM sits within a family of **dominant-negative keratin disorders**. Pachyonychia congenita — including the K17/PC-K17 subtype allelic with SM — is *"an autosomal dominant genodermatosis caused by heterozygous mutations in any one of the genes encoding the differentiation-specific keratins K6a, K6b, K16, or K17,"* with epidermal cysts among its features (PMID: 21430705). The dominant-negative aggregation mechanism is directly druggable: Leachman et al. developed a mutant allele-specific siRNA (TD101) against KRT6A N171K and demonstrated *"potent inhibition of the mutant allele and reversal of the cellular aggregation phenotype"* in vitro (PMID: 18495438), then advanced it to a first-in-human phase Ib trial: *"This trial represents the first time that siRNA has been used in a clinical setting to target a mutant gene or a genetic disorder, and the first use of siRNA in human skin"* (PMID: 19935778), achieving callus regression without adverse events. Allele-specific RNAi can discriminate single-nucleotide differences, silencing only the mutant allele (PMID: 26264709). This provides a clear, mechanism-matched therapeutic avenue for SM.

Finding 7 — KRT17 mutations specifically associate with cysts and natal teeth; hotspot mutations destabilize keratin assembly (structural modeling)

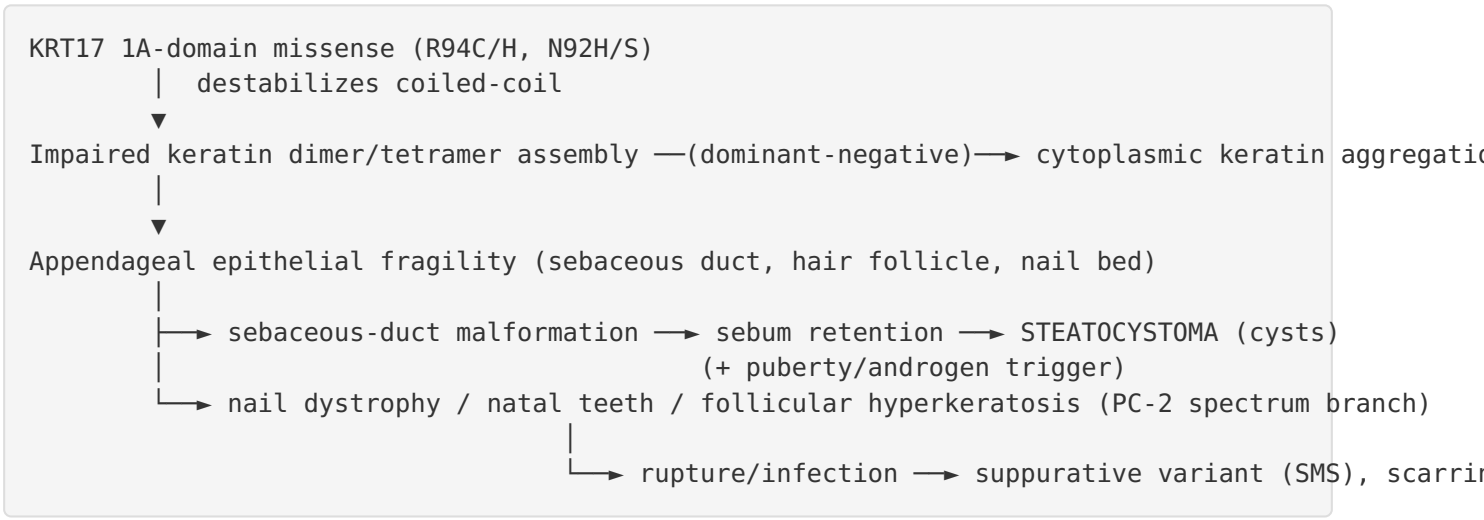
Within genetically confirmed pachyonychia congenita cohorts, the KRT17 genotype is specifically associated with cysts. In 254 PC patients, Eliason et al. found *"a strong association of natal teeth and cysts in carriers of a KRT17 mutation"* (PMID: 22264670). Wu et al. integrated genotype, protein structure, and phenotype using the International PC Research Registry, confirming *"cutaneous cysts, follicular hyperkeratosis, and natal teeth with K17 mutations,"* and provided the structural mechanism: *"hotspot missense mutations in PC largely disrupted hydrophobic interactions or surface charge. The former may destabilize keratin dimers/tetramers, whereas the latter likely interferes with higher-order keratin filament formation"* (PMID: 34116063). Specific KRT17 alleles further modify phenotype; e.g., KRT17 p.L99P is associated with increased fingernail involvement and 20-nail dystrophy, whereas p.N92S trends milder (PMID: 33486795). This explains why cysts are the KRT17-signature manifestation and links the Arg94/Asn92 hotspots mechanistically to impaired filament assembly.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous missense mutation** arises in the 1A helix-initiation rod domain of *KRT17* (hotspots R94C/R94H, N92H/N92S) — germline, inherited autosomal dominant (or de novo). *(Demonstrated by co-segregation studies.)*
2. The mutation **leads to** a single amino-acid substitution at a conserved residue critical for coiled-coil dimer formation. *(Demonstrated.)*
3. The altered residue **disrupts** hydrophobic packing and/or surface-charge interactions needed for keratin heterodimer and tetramer assembly (K17 pairs with a type II partner keratin). *(Inferred from protein-structure modeling, Wu 2021.)*
4. Because the mutant protein is still expressed and incorporated, it **acts in a dominant-negative fashion**, poisoning the intermediate-filament network assembled from both wild-type and mutant subunits. *(Inferred from the keratinopathy paradigm; directly demonstrated for homologous keratins via aggregation-reversal siRNA experiments.)*

5. Defective assembly **results in** cytoplasmic keratin aggregation and mechanical/stress fragility specifically in **K17-expressing appendageal epithelia** (sebaceous duct, hair follicle, nail bed). *(Inferred; aggregation demonstrated in homologous K6a system.)*
6. In the **sebaceous follicle/duct**, compromised epithelial integrity and abnormal keratinization **malform the sebaceous duct**, which **branches** toward: (a) **sebum retention** within a closed, agranular squamous-lined cyst → the steatocystoma lesion; and (b) **impaired appendageal homeostasis** manifesting (variably) as nail dystrophy, follicular hyperkeratosis, and natal teeth on the PC spectrum. *(Lesion histology demonstrated; branch to PC features demonstrated in allelic patients.)*
7. Androgenic activation of sebaceous glands around **puberty leads to** clinical emergence of multiple cysts (explaining the early-twenties onset). *(Inferred from onset timing and sebaceous-gland biology.)*
8. Secondary events — rupture, bacterial infection, inflammation — **result in** the suppurative variant (SMS), scarring, and occasional overlap with hidradenitis suppurativa. *(Demonstrated clinically.)*



Upstream vs downstream

Level	Element	Evidence type
Upstream (initiating)	KRT17 1A-domain missense mutation (R94, N92)	Human genetic (co-segregation)
Molecular	Loss of dimer/tetramer stability; dominant-negative filament poisoning	Structural modeling + keratinopathy paradigm
Cellular	Keratin aggregation, appendageal epithelial fragility	In vitro (homologous keratins)
Tissue	Sebaceous-duct malformation, sebum retention cyst	Human histopathology
Clinical (downstream)	Multiple cysts; variable nail/tooth features; suppuration	Human clinical

Ontology term suggestions

- **Gene/protein:** KRT17 (HGNC:6427); GO:0045095 keratin filament; GO:0005882 intermediate filament; GO:0030280 structural constituent of skin epidermis; GO:0008544 epidermis development; GO:0045109 intermediate filament organization.
- **Cell types (CL):** CL:0000312 keratinocyte; CL:0002190 sebaceous gland cell (sebocyte); CL:0002559 hair follicle cell.
- **Anatomy (UBERON):** UBERON:0002073 hair follicle; UBERON:0001821 sebaceous gland; UBERON:0002067 dermis; UBERON:0001003 skin epidermis; UBERON:0000014 zone of skin (trunk/axilla/groin).
- **Phenotype (HPO):** HP:0032320 cutaneous cyst/steatocystoma; HP:0200039 papule; HP:0001596 alopecia (K17 biology); HP:0008069 natal tooth; HP:0008404 nail dystrophy.
- **Disease (MONDO):** MONDO:0008485 steatocystoma multiplex.
- **Chemical (CHEBI):** CHEBI:6716 isotretinoin (treatment); sebum constituents (triglycerides, squalene, wax esters).
- **Treatment (NCIT):** surgical excision; isotretinoin therapy; cryosurgery; laser therapy.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
9008238	K17 missense mutations → PC-2 or SM-like phenotype	Human genetic	F1 — N92H/R94H in SM families
9767294	K17 mutations cause SM or PC-2	Human genetic	F1, F2 — R94C/N92S; allelism; mutation-independent phenotype
26165312	R94C in SM	Human genetic	F1 — co-segregation in 4-gen pedigree
35676340	Recessive atypical PC (KRT17 R94H)	Human genetic	F2 — dominant vs recessive KRT17 spectrum
16702408	K17 modulates hair cycling (TNFα)	Mouse model	F3 — K17 appendageal biology
33008845	K17 regulates nuclear morphology/chromatin	In vitro	F3 — non-structural K17 functions
41880216	K17 in Rac1-mediated DDR	In vitro	F3 — K17 stress response
41363957	Steatocystomas: update	Clinical review	F4 — core lesion definition
33088817	SM case (anti-IL-12/23)	Human clinical	F4 — distribution/asymptomatic
29998099	Facial SM clinicopathology	Human clinical	F4 — histopathology
42518883	Epidemiology systematic review	Aggregated (145 cases)	F5 — onset, familial, hotspot
21430705	PC molecular genetics	Review/ registry	F6 — dominant-negative keratin class
18495438	Therapeutic siRNA for PC	In vitro	F6 — aggregation reversal
19935778	First-in-human siRNA trial	Clinical trial (Ib)	F6 — allele-specific siRNA validated
26264709	Allele-specific RNAi review	Review	F6 — single-nucleotide discrimination
22264670	254 PC patients phenotype	Human cohort	F7 — cysts/natal teeth ↔ KRT17
34116063			F7 — structural mechanism

PMID	Title (abbrev.)	Evidence type	Supports
	Genotype–structurotype–phenotype	Human + modeling	
33486795	Predictive genetic variants in PC	Human cohort	F7 — allele-specific modifiers

Supporting context papers: K17 in disease pathogenesis (psoriasis to dermatoses) (PMID: 30306595); PC-associated PPK pathophysiology and Keap1-Nrf2 (PMID: 31021398); PC clinical features and future treatments (PMID: 37766547); molecular clinical endpoints for siRNA trials (PMID: 21191405); treatment case reports — isotretinoin + cryotherapy for SMS (PMID: 10812703), combination laser therapy (PMID: 22487444), SMS review (PMID: 41736170), hybrid SM/EVHC cyst (PMID: 40998547).

Report by Template Section

1. Disease Information

Steatocystoma multiplex is a rare, benign, autosomal dominant genodermatosis of the pilosebaceous unit, characterized by multiple sebum-filled dermal cysts. **Identifiers:** MONDO:0008485; OMIM 184500; causal gene *KRT17* (HGNC:6427; NCBI Gene 3872); Orphanet (steatocystoma multiplex); ICD-10 L72.2 (steatocystoma multiplex); MeSH "Steatocystoma Multiplex." **Synonyms:** sebocystomatosis; steatocystoma multiplex (familial). Information is derived from **aggregated disease-level resources** (OMIM, Orphanet, systematic review) and from individual case reports/pedigrees rather than EHR-scale cohorts.

2. Etiology

Primary cause: genetic — heterozygous missense mutations in the 1A domain of *KRT17* (dominant-negative). **Genetic risk:** causal hotspot variants at Arg94 (R94C, R94H) and Asn92 (N92H, N92S); a positive family history is the dominant risk factor (familial cases ~26%). **Environmental/hormonal:** androgen-driven sebaceous activation at puberty is the main modulator of lesion emergence; no toxins, infections, or occupational exposures are established as causal. **Protective factors:** none genetically or environmentally established.

Gene–environment interaction: the mutation-independent SM-versus-PC2 phenotype (identical R94C → different disease) implies modifier genes and/or hormonal/environmental context determine phenotypic expression (PMID: 9767294).

3. Phenotypes

- **Multiple cutaneous cysts** (physical manifestation): firm, non-tender, skin-colored to yellowish dermal papules/nodules; onset adolescence/early adulthood (median 23 y); slow/chronic progression; defining, near-universal feature. HPO: HP:0032320 (cutaneous cyst/steatocystoma); HP:0200039 (papule).
- **Sebum discharge** on puncture (clinical sign).
- **Suppuration/secondary infection** (SMS variant): episodic inflammation, pain, scarring; higher in familial cases.
- **Nail dystrophy** (variable; PC spectrum overlap): HP:0008404.
- **Natal teeth / dental anomalies** (variable, KRT17-associated): HP:0008069.
- **Quality-of-life impact:** usually asymptomatic but can be *"significantly disfiguring"* (PMID: 31915618); psychosocial burden, particularly with facial/extensive involvement; coil-1A mutation carriers report more emotional issues (Wu 2021). Does not affect lifespan.

4. Genetic / Molecular Information

Causal gene: *KRT17* (gene OMIM 148069), type I intermediate filament, chromosome 17q21.2. **Pathogenic variants:** missense substitutions in the 1A helix-initiation motif — R94C (c.280C>T), R94H, N92H, N92S; classified pathogenic/likely pathogenic by co-segregation and recurrence. **Allele frequency:** absent/extremely rare in population databases (gnomAD), consistent with high-penetrance dominant disease. **Origin:** germline. **Functional consequence:** dominant-negative (not simple loss of function) — mutant subunit impairs filament assembly (PMID: 34116063). **Modifier genes:** unidentified but inferred from mutation-independent SM/PC2 divergence and allele-specific nail severity (e.g., KRT17 L99P) (PMID: 33486795). **Epigenetics / chromosomal abnormalities:** none established for SM.

5. Environmental Information

No environmental toxins, radiation, pollution, occupational exposures, lifestyle factors, or infectious agents are established as causes. Hormonal (androgenic) drive at puberty modulates expression. Secondary bacterial infection contributes to the suppurative variant but is a downstream complication, not a cause.

6. Mechanism / Pathophysiology

See the **Mechanistic Model** section above for the full ordered causal chain. In brief: KRT17 1A-domain mutation → destabilized keratin dimer/tetramer assembly → dominant-negative filament poisoning and cytoplasmic aggregation in K17-expressing appendageal epithelia → sebaceous-duct malformation → sebum-retention cysts, with a branch to variable nail/tooth features on the PC spectrum and a secondary branch to suppuration/scarring. No single canonical signaling pathway (Wnt/MAPK/mTOR) is causal; relevant K17-linked signaling includes TNFα/NF-κB (hair cycling) and Keap1-Nrf2/oxidative-stress responses described in the broader PC literature.

7. Anatomical Structures Affected

Primary organ: skin (integumentary system). **Specific sites:** pilosebaceous unit — sebaceous gland/duct (UBERON:0001821), hair follicle (UBERON:0002073), within the dermis (UBERON:0002067). **Distribution:** trunk, axillae, groin, limbs (occasionally face/scalp; rare acral subcutaneous variant). **Tissue type:** epithelial (stratified squamous, agranular cyst lining). **Cells:** sebocytes (CL:0002190), keratinocytes (CL:0000312). **Subcellular:** cytoplasmic intermediate-filament/keratin network (GO:0045095 keratin filament; GO:0005856 cytoskeleton). **Lateralization:** bilateral/generalized, non-lateralized.

8. Temporal Development

Onset: typically peri-/post-pubertal; median onset 23 years (familial 13.5 y, sporadic 30.0 y) (PMID: 42518883). **Pattern:** insidious, chronic. **Progression:** slow accumulation of lesions; stable-to-progressive; lifelong. Suppurative variant is episodic/relapsing. **Remission:** no spontaneous resolution; individual lesions can be cleared by procedures. **Critical window:** puberty (androgen-driven sebaceous activation) marks clinical emergence.

9. Inheritance and Population

Inheritance: autosomal dominant (rare homozygous KRT17 → recessive atypical PC, PMID: 35676340). **Penetrance:** high, with variable expressivity. **Prevalence:** rare; no precise figure; no clear geographic/ethnic predilection (PMID: 42518883). **Sex ratio:** roughly equal overall; HS co-occurrence female-predominant (12.5% vs 1.5%, P=0.019). **Familial fraction:** ~26.2% of reported cases; remainder sporadic (de novo or under-ascertained). **Anticipation / founder effects / consanguinity:** not established (recurrent R94 reflects a mutational hotspot, not a founder haplotype).

10. Diagnostics

Clinical: recognition of multiple characteristic cysts; puncture reveals oily/sebum material. **Histopathology (gold standard):** dermal cyst with agranular stratified squamous lining, eosinophilic corrugated cuticle, and sebaceous lobules in the wall (PMID: 29998099). **Dermoscopy:** yellow homogeneous background with peripheral linear vessels. **Genetic testing:** single-gene KRT17 sequencing (exon 1 / 1A domain) or keratin/PC gene panel confirms diagnosis and distinguishes mimics; WES/WGS useful in atypical cases. **Differential diagnosis:** eruptive vellus hair cysts (can co-occur/form hybrid cysts, PMID: 40998547), epidermoid cysts, milia, hidradenitis suppurativa (for SMS), trichofolliculoma. No specific blood biomarker or imaging test.

11. Outcome / Prognosis

Benign; no malignant potential; normal life expectancy. Morbidity is cosmetic and psychosocial; the suppurative variant causes pain, recurrent infection, and scarring. Lesions are chronic and do not spontaneously remit. Prognostic factors for severity include familial disease and early onset (associated with infected lesions and nail dystrophy). No disease-specific mortality.

12. Treatment

No cure; symptom-directed. **Pharmacotherapy:** oral isotretinoin (CHEBI:6716; reduces sebaceous activity/inflammation, especially for SMS — PMID: 10812703); antibiotics for secondary infection; biologics (anti-TNF, anti-IL-12/23, anti-IL-17) reported for suppurative/HS-overlap disease (PMID: 41736170). **Procedural:** surgical excision (most definitive for isolated lesions), incision/drainage, mini-incision extraction, cryotherapy, and laser (1450-nm diode targeting sebaceous glands + 1550-nm fractional erbium targeting cysts — PMID: 22487444). **Emerging/experimental:** mutation-targeted **allele-specific siRNA** (paradigm validated in PC: TD101 — PMID: 19935778); no SM-specific approved targeted therapy yet. NCIT terms: surgical excision, isotretinoin therapy, cryosurgery, laser therapy.

13. Prevention

No primary prevention (genetic etiology). **Secondary/tertiary:** early recognition to shorten the ~8-year diagnostic delay; treatment of individual lesions and prompt management of suppuration to prevent scarring. **Genetic counseling:** autosomal dominant → 50%

transmission risk; offer KRT17 testing and family counseling; prenatal/preimplantation testing is technically feasible but rarely pursued given the benign, non-lethal nature. No vaccine or public-health intervention applicable.

14. Other Species / Natural Disease

Human disease (NCBI Taxon 9606). Orthologous gene *Krt17* exists in mouse (NCBI Gene 16667) and other mammals. **Krt17-null mice** develop alopecia and hair-shaft fragility (PMID: 16702408), modeling appendageal K17 function but **not** the human cystic phenotype (null vs dominant-negative). No naturally occurring SM-equivalent disease is documented in companion animals or wildlife (OMIA); no zoonotic relevance.

15. Model Organisms

Mouse: Krt17 knockout (alopecia/hair-cycle phenotype) recapitulates appendageal dependence on K17 but not cysts; the dominant-negative human mechanism is not reproduced by a null allele. A dominant-negative R94 knock-in would be the ideal, currently unavailable, model. **In vitro:** patient-derived keratinocytes and transfected cell systems demonstrate keratin aggregation (shown for homologous keratins). **iPSC/organoid** sebaceous or skin-appendage models are plausible future systems. Resources: MGI (Krt17), IMPC/IMSR. **Limitations:** no model fully recapitulates the human sebaceous-duct cyst; most mechanistic inference is extrapolated from homologous PC keratins (K6a/K16) and structural modeling.

Limitations and Knowledge Gaps

1. **No dominant-negative animal model of SM.** Available mouse data come from Krt17-null (loss-of-function) animals, which model K17's role in hair cycling but not the human dominant-negative cystogenesis; the sebaceous-duct cyst phenotype has not been recapitulated in vivo.
2. **Mechanism partly inferred by homology.** Direct demonstration of mutant K17 aggregation and filament poisoning for the specific SM hotspot alleles is limited; aggregation-reversal and allele-specific-siRNA evidence derives from homologous keratins (K6a) in PC.
3. **Modifier genes unidentified.** The mutation-independent SM-versus-PC2 divergence strongly implies genetic/environmental modifiers, but none are molecularly identified.

4. **Epidemiology is case-report-based.** Even the systematic review aggregates only 145 cases; true prevalence/incidence, penetrance estimates, and population-level data are lacking, with likely publication/ascertainment bias toward severe and familial cases.
5. **No controlled treatment trials.** Therapeutic evidence is case reports/series; no randomized data exist for isotretinoin, laser, or biologics in SM, and allele-specific siRNA has not been tested in SM specifically.
6. **Androgen/puberty link is inferential.** The onset timing suggests hormonal modulation, but direct mechanistic data in SM are absent.

Proposed Follow-up Experiments / Actions

1. **Generate a Krt17 R94C/H knock-in mouse** (dominant-negative) and characterize sebaceous-duct architecture, cyst formation, and androgen responsiveness to create the first faithful SM model.
2. **Patient-derived sebocyte/skin organoids or iPSC appendage models** carrying SM hotspot alleles to test keratin aggregation and filament assembly directly in the relevant cell type.
3. **Allele-specific siRNA/ASO against KRT17 R94C** — design and validate single-nucleotide-discriminating knockdown in patient keratinocytes/sebocytes, then intralesional proof-of-concept, mirroring the TD101 PC paradigm.
4. **Modifier-gene discovery** via whole-genome sequencing and phenotype association in SM-versus-PC2 families sharing identical KRT17 alleles.
5. **Prospective natural-history registry** to establish penetrance, prevalence, sex ratio, QoL metrics (DLQI/SF-36), and standardized severity scoring — reducing reliance on case reports.
6. **Structural/biophysical validation** (cryo-EM, in vitro filament-assembly assays) of how R94 and N92 substitutions specifically destabilize K17 dimer/tetramer formation.
7. **Clinical trial design** for isotretinoin dosing and laser combinations (1450-nm + 1550-nm) with objective lesion-count endpoints.

Consensus Answer

Steatocystoma multiplex (MONDO:0008485; OMIM 184500) is an autosomal dominant benign genodermatosis caused by heterozygous missense mutations in the 1A helix-initiation rod domain of **KRT17 (keratin 17)**, with Arg94 and Asn92 as recurrent hotspots. These mutations act by a dominant-negative mechanism—destabilizing keratin dimer/tetramer assembly and causing cytoplasmic keratin aggregation—that malforms the sebaceous-follicle duct and produces multiple sebum-retention cysts, typically emerging in the early-to-mid twenties. SM is allelic with pachyonychia congenita type 2 on a single KRT17 phenotypic spectrum (cysts being the KRT17 signature); it does not affect lifespan, and management is symptomatic (excision, drainage, isotretinoin, laser, biologics for the suppurative variant), with allele-specific siRNA as a rational emerging mechanism-targeted therapy.